Project Methodology

Contents

Chosen Methodology: Kanban

Justification:
The Generic Demand and Capacity Modelling Platform for NHFT Services is a data-driven, exploratory project with ongoing development, testing, and refinement phases. The work involves continuous iteration across data pipelines, model tuning, dashboard development, and stakeholder validation.

Kanban is the most suitable methodology because:

- It allows flexible task management across multiple workstreams (data engineering, modelling, and visualisation).
- Progress can be visually tracked using Kanban boards (aligned with NHFT BI's existing workflow).
- Tasks can be prioritised dynamically as stakeholder feedback and data issues arise.
- It supports incremental delivery, where functional components (e.g., forecasting module, simulation engine, dashboard) are completed and demonstrated continuously rather than waiting for final integration.
- It encourages continuous improvement and collaboration across BI and Performance Improvement teams.

This approach aligns with NHFT's culture of iterative improvement, allowing progress to be monitored transparently while maintaining flexibility for mid-project refinements.

Although the project follows a largely linear progression from documentation and governance, through data access, modelling, and application development, it will be managed using Microsoft Planner in a Kanban-style workflow.

Method	Strengths	Limitations	Fit for This Project
Waterfall	Clear, linear, milestone-driven; good for fixed requirements.	Inflexible; difficult to adjust if data issues or new insights arise.	Useful for overall high-level milestone structure, but too rigid for modelling and analytical iteration.
Agile (Scrum)	Sprint-based, collaborative, regular checkpoints.	Heavy ceremony overhead; requires multi-person team structure.	Helpful conceptually but not ideal for a single-developer project.
Kanban	Highly flexible; visual task flow; suitable for continuous delivery and parallel workstreams; supports dynamic reprioritisation.	Requires strong self-discipline to maintain WIP limits.	**Best fit**: aligns with BI working practices, supports modelling iteration, and integrates well with Microsoft Planner.

Project Timeline

The linear structure is reflected in the numbered overarching tasks and deliverables, but Microsoft Planner provides the flexibility to organise, prioritise, and monitor work at the sub-task level, ensuring transparency and continuous progress.

Workstream / Milestone	Dec 2025 (Weeks 1–4)	Jan 2026 (Weeks 1–4)	Feb 2026 (Weeks 1–4)	Deliverable / Outcome
1. Project Setup & Planning				Project Agreement confirmed; Kanban board created; stakeholder sign-off.
2. Data Access & Governance				Secure data connections to NHFT SQL warehouse; DPIA approved; governance confirmed.
3. Data Engineering & Automation				Automated data pipelines and validation scripts created; direct-query model established.
4. Forecasting Model Development				Time-series and ML forecasting modules implemented and tested.
5. Capacity & Simulation Modelling				Simulation and queueing components integrated into framework.
6. Dash Application Development				Interactive visualisation interface developed using Dash; initial pilot view deployed.
7. Testing, Validation & Feedback				Forecast validation, stakeholder feedback, and iteration rounds completed.
8. Documentation & Handover				Technical documentation, configuration templates, and OneNote log updates finalised.

The Planner board will include the following Kanban columns:

- **Backlog:** All planned and future tasks, grouped by overarching subjects.
- **Up Next:** Prioritised work items prepared for immediate action.
- **In Progress:** Tasks actively being developed or implemented.
- **In Review:** Work undergoing testing, validation, or stakeholder feedback.
- **Issues:** Any blocked items requiring clarification, rework, or risk mitigation.
- **Completed:** Tasks that meet acceptance criteria and are formally signed off.

Each overarching task (e.g., *Data Access*, *Forecasting Module*, *Simulation Engine*, *Dash App*) will contain subtasks organised using Microsoft Planner checklists.

To support planning and workload management, Fibonacci effort estimates (1, 2, 3, 5, 8, 13…) will be assigned to subtasks. This helps approximate complexity and time investment more realistically than fixed time estimates.

User Stories

As of today, the user stories for this project are:

1. Project Documentation & Governance

Completion of project initiation and governance documents required for NHFT and academic approval.

Due: 10/12/2025

1.1 Project Agreement (2) - 15/10/2025
1.2 Ethics & Legal Assessment (3) - 05/11/2025
1.3 Data Protection Impact Assessment (DPIA) (5) - 10/12/2025
1.4 Project Approach (methodology + justification) (2) - 19/11/2025
1.5 Project Plan (Gantt/Kanban breakdown) (3) - 10/12/2025
1.6 Stakeholder Summary & RACI Matrix (3) - 10/12/2025

2. Data Access & Infrastructure Setup

Securing data access, BI environment connections, and ensuring IG compliance for operational datasets.

Due: 15/12/2025

2.1 Submit BI Data Access Request (2)
2.2 IG / Data Governance Approval (3)
2.3 Validate SQL Connections to NHFT BI Warehouse (3)
2.4 Set Up Secure Python Environment (Anaconda / Venv) (2)
2.5 Create Version Control Repo (GitHub) (1)
2.6 Build Folder Structure for Project Code & Outputs (1)

3. Data Engineering & Automation Pipeline

Build automated SQL-driven data ingestion pipelines with validation and pseudonymisation.

Due: 09/01/2026

3.1 Identify Required Tables & Fields (Referrals/Appts/Workforce) (3)

3.2 Build SQL Queries for Pilot (3 services) (5)
3.3 Implement Python Data Extraction Module (5)
3.4 Develop Unit Tests, Data Quality & Validation Scripts (5)
3.5 Create Automated Direct-Query Pipeline (No file storage) (8)
3.6 Document Data Dictionary & Metadata (3)

4. Forecasting Module (MVP)

Develop forecasting engine for demand using statistical and ML models, supporting three pilot services.

Due: 06/02/2026

4.1 Exploratory Time-Series Analysis for 3 Services (5)
4.2 Implement Core Forecast Models (ARIMA, ETS, XGBoost) (8)
4.3 Build Model Selection Logic (lowest error metrics) (8)
4.4 Add Fairness/Balance Metrics Across Services (5)
4.5 Generate Forecast Visuals for Validation (3)
4.6 Document Forecasting Methods & Results (3)

5. Capacity & Simulation Module

Add simulation and queueing components to model staff availability, sessions, rooms, and DNA rates.

Due: 13/02/2026

5.1 Define Capacity Parameters by Service Type (templates) (5)
5.2 Implement Queueing Models (M/M/s or M/G/s) (8)
5.3 Build Simulation Engine Using SimPy (13)
5.4 Add Scenario Functions (staffing, demand variation, DNA rates) (5)
5.5 Validate Simulation Outputs with Historical Data (5)
5.6 Document Capacity & Simulation Logic (3)

6. Dash / Power BI Application

Build interactive dashboards to visualise forecasting performance, breaches, utilisation, and scenarios.

Due: 27/02/2026

6.1 Create Wireframe for Dashboard Structure (3)
6.2 Build Dash Front-End Layout (Tabs/Widgets) (5)
6.3 Integrate Forecasting Outputs into Dash (5)
6.4 Integrate Capacity & Simulation Outputs (8)
6.5 Add Scenario Controls (sliders, toggles, dropdowns) (5)
6.6 User Testing With Pilot Service Managers (3)
6.7 Apply Styling, Tooltips, & Final UI Enhancements (2)

7. Testing, Validation & Refinement

Ensure model accuracy, handle exceptions, tune parameters, refine the dashboard, and quality-check outputs.

Due: 27/02/2026

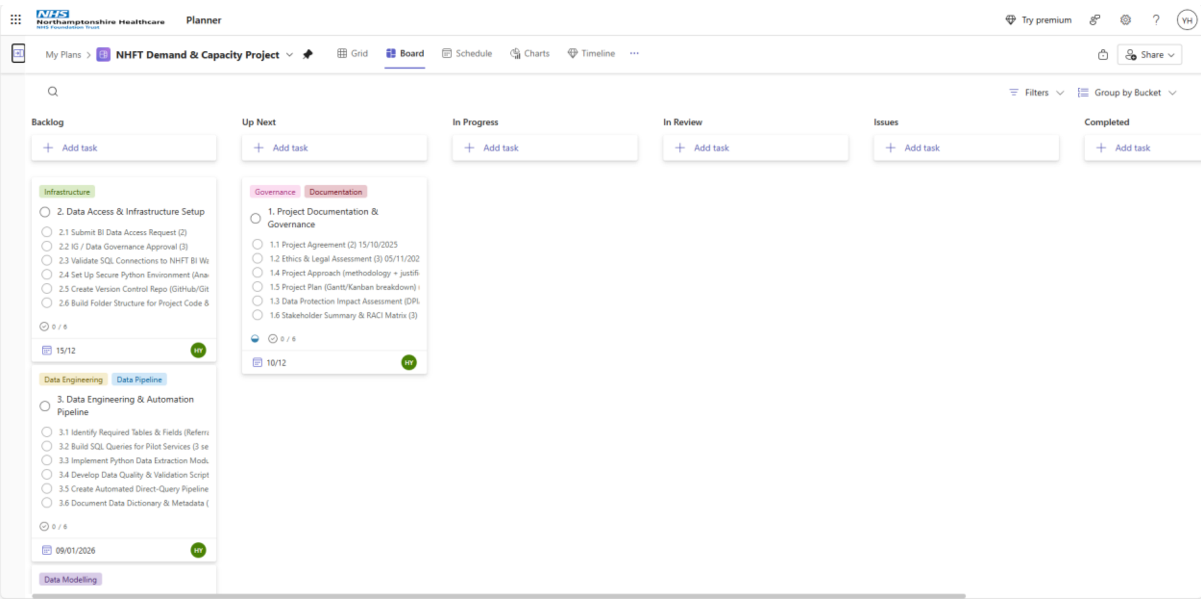
7.1 Compare Forecasts Against Actuals (RMSE, MAE) (5)
7.2 Parameter Tuning & Fairness Adjustments (5)
7.3 Stress-Test Direct-Query Pipeline (3)
7.4 Resolve Model Errors & Outlier Handling (5)
7.5 Conduct Feedback Session With Stakeholders (3)
7.6 Final Corrections Before MVP Sign-Off (2)

8. Documentation, Logbook & Final Presentation Prep

Compile project documentation, update OneNote log, record presentation, and finalise deliverables for EPA.

Due: 01/06/2026

8.1 Update OneNote Log with Reflections (1)
8.2 Prepare Final Written Report / Portfolio Evidence (5)
8.3 Create Presentation Slides (3)
8.4 Record Pitch Video (2)
8.5 Present to Stakeholders (2)



Summary Reflection

Selecting the right methodology ensures that the project's scope, resources, and outcomes remain aligned with organisational needs. In my work-based project, Kanban supports iterative progress, ongoing collaboration, and effective time management. This complements the analytical nature of the project, enabling a steady, evidence-based evolution of the Demand and Capacity Modelling Platform.